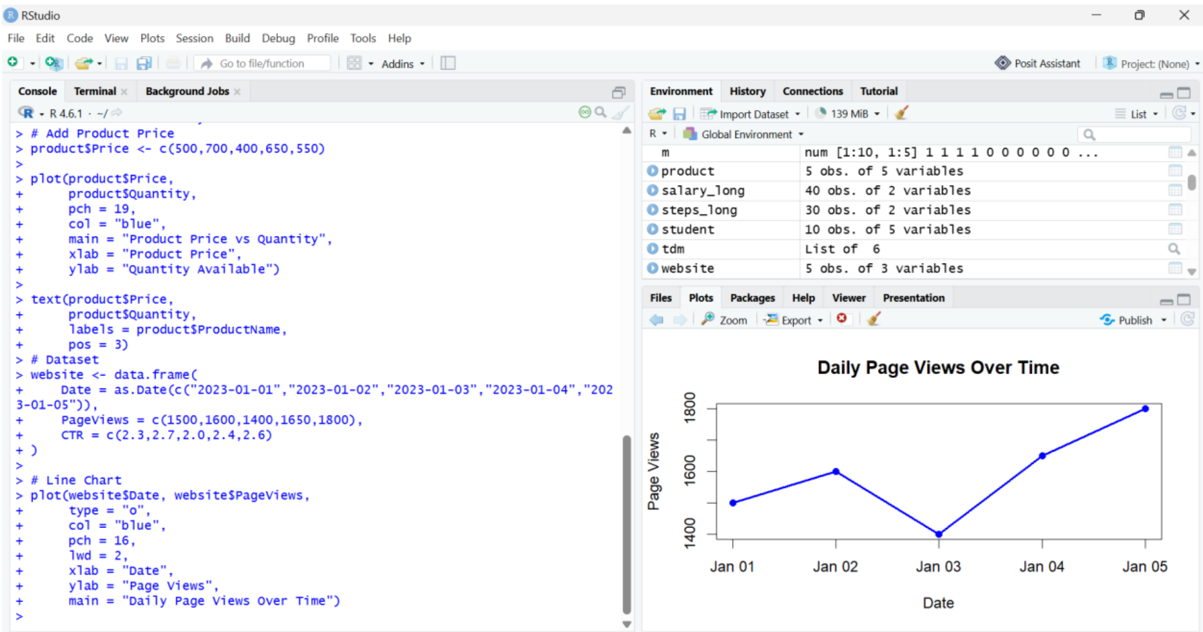
lab 5: Website Analytics

Question 1: Line Chart (Daily Page Views)

Code:

```
website <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6)
)
plot(website$Date, website$PageViews,
     type = "o",
     col = "blue",
     pch = 16,
     lwd = 2,
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views Over Time")
```

Output:

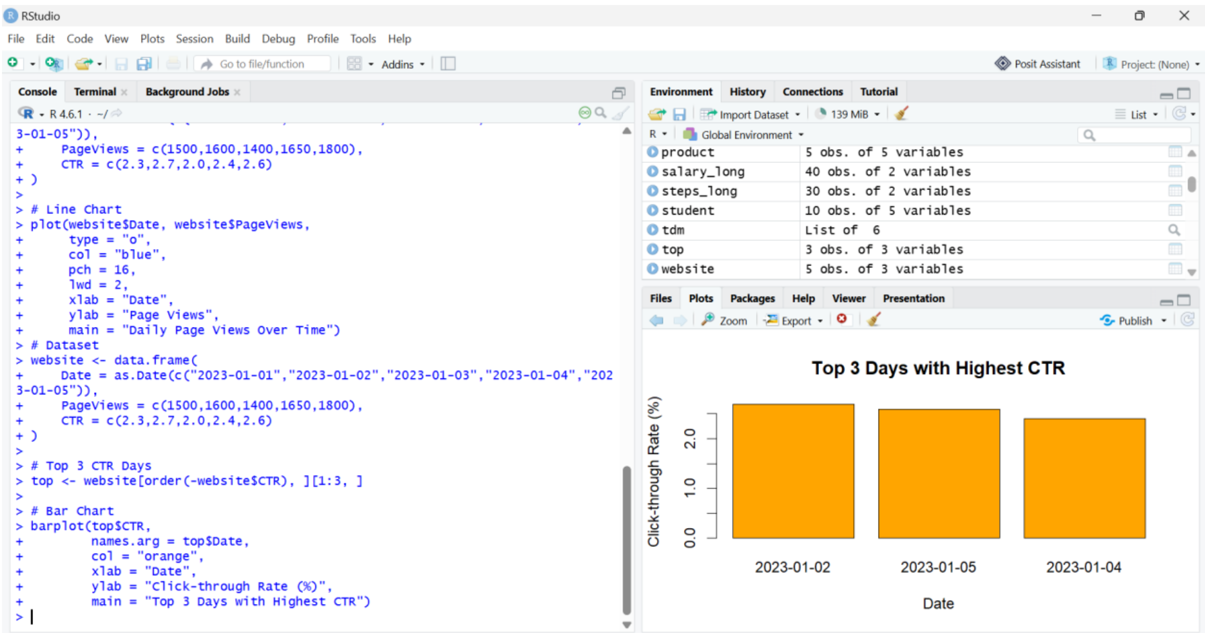


Question 2: Bar Chart (Top 3 Days with Highest CTR)

Code:

```
website <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6))
top <- website[order(-website$CTR), ][1:3, ]
barplot(top$CTR,
        names.arg = top$Date,
        col = "orange",
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top 3 Days with Highest CTR")
```

Output:



Question 3: Stacked Area Chart (Likes, Shares, Comments)

Code:

```
date <- as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05"))
likes <- c(300,350,280,370,400)
shares <- c(120,150,100,140,160)
```

```
comments <- c(80,90,70,85,95)
library(plotrix)
stackpoly(date,
          cbind(likes, shares, comments),
          col = c("skyblue","lightgreen","pink"),
          xlab = "Date",
          ylab = "User Interactions",
          main = "Stacked Area Chart")
legend("topleft",
       legend = c("Likes","Shares","Comments"),
       fill = c("skyblue","lightgreen","pink"))
```

Output: